From a Fractional Integral to the Frontier of the Riemann Hypothesis

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Abstract

We revisit a representation of the Riemann zeta function through the Weyl fractional integral of the geometric series of exponentials, prove that it is equivalent to the classical Mellin representation, and compare it with the finite-base-point Riemann–Liouville and Caputo variants, whose endpoint terms preclude the exact eigenrelation the construction requires. Using this gateway as an organizing thread, we give an expository traversal of the analytic theory surrounding the Riemann Hypothesis: the reformulation of the Hypothesis as the reality of the zeros of a strip transform; a precise account of why arguments using only the functional equation and reflection symmetry cannot locate zeros, with the Davenport–Heilbronn function as the governing obstruction; the Riemann–Weil explicit formula and the positivity criteria of Weil and Li, including a self-contained proof of both directions of Li's criterion in which the key difficulty is dissolved by the geometry of a Cayley transform; an identification of the fractional semigroup law with the Mellin translation flow, together with a discussion of its limitations for zero location; and a survey of the unconditional estimation ladder (Hardy's theorem, the approximate functional equation, the second and fourth moments, the Bohr–Landau theorem), including a compact proof that the number of zeros on the critical line up to height T exceeds $T^{3/4}/\log^5 T$. The paper is expository: its principal theorems are classical and attributed throughout; a small number of observations and arrangements may be of independent interest and are identified as such.

1 Introduction

Let $\zeta(s) = \sum_{k\geq 1} k^{-s}$ for $\mathrm{Re}(s) > 1$, continued meromorphically to $\mathbb{C}$ with a simple pole at $s = 1$. The Riemann Hypothesis (RH) asserts that every non-real zero of ζ has real part $\frac{1}{2}$.

The starting point of this paper is an identity from the author's 2021 notebooks. Writing $D^{-\alpha}$ for the fractional *integration* operator normalized so that $D^{-\alpha}[e^{kx}] = k^{-\alpha}e^{kx}$ (the notebooks' convention writes this as a fractional derivative of order $-\alpha$), a telescoping computation with the geometric series gives, formally,

$$\lim_{x\to 0^-} \lim_{N\to\infty} D^{-\alpha}\left[\frac{e^x\big(1-e^{Nx}\big)}{1-e^x}\right] \;=\; \sum_{k=1}^{\infty}\frac{1}{k^{\alpha}} \;=\; \zeta(\alpha), \tag{1}$$

with the plan of using the fractional representation as an alternate version of ζ from which to study the Hypothesis. Section 2 makes (1) rigorous, identifies the operator (it is the Weyl fractional *integral*), proves that the identity is equivalent to the classical Mellin representation $\Gamma(\alpha)\zeta(\alpha) = \int_0^\infty t^{\alpha-1}(e^t-1)^{-1}dt$, and compares it with the finite-base-point Riemann–Liouville and Caputo variants, whose endpoint memory precludes the exact eigenrelation; that relation is recovered exactly in the translation-invariant (base point $-\infty$) setting, of which the Weyl operator is the instance used here.

The rest of the paper follows the gateway to its natural destinations, with one standard maintained throughout: every principal statement is either proved in full or imported with a precise citation, and each section closes by locating its content relative to the open problem. The itinerary: Section 3 reformulates RH as the reality of the zeros of a strip transform and proves exactly why symmetry alone cannot decide it; Section 4 presents the explicit formula and the positivity criteria, with a complete proof of Li's criterion; Section 5 identifies the distinctive algebraic structure of the fractional formulation — its semigroup law — with the Mellin translation flow and discusses its limitations for zero location; Section 6 surveys the unconditional estimation ladder and proves two of its statements in full; Section 7 collects the structural conclusions; and Section 8 closes with the Robin–Lagarias inequality.

On the nature of this paper. The theorems of Sections 4 and 6 are classical — Weil [26], Li [18], Hardy [9], Hardy–Littlewood [10, 11], Ingham [12], Bohr–Landau, Selberg [24], Levinson [17], Conrey [5] — and no novelty is claimed for them. What may be of independent interest is the arrangement: the equivalence and operator-comparison statements of Section 2; the two-point analysis of Proposition 3.2; the presentation of Li's criterion in which the extremal difficulty of Weil-type converses is dissolved by compactification (Remark 4.6); the semigroup identification of Section 5; the diagnostic observation of Remark 6.1; and the short proof of Theorem 6.2.

2 The fractional gateway

Definition 2.1. For $\mathrm{Re}(\alpha) > 0$ and g measurable and locally bounded on $[0,\infty)$ with $g(t) = O(e^{-ct})$ as $t \to +\infty$ for some $c > 0$ — hypotheses under which the displayed integral converges absolutely for every $x \geq 0$ — the *Weyl fractional integral* of order α is

$$(W^{-\alpha}g)(x) := \frac{1}{\Gamma(\alpha)} \int_x^\infty (t-x)^{\alpha-1} g(t)\, dt, \qquad x \geq 0.$$

On this class, and with $\mathrm{Re}\alpha, \mathrm{Re}\beta > 0$, Fubini's theorem gives the semigroup law $W^{-\alpha}W^{-\beta} = W^{-(\alpha+\beta)}$.

Lemma 2.2. *For every integer $k \geq 1$ and $\mathrm{Re}(\alpha) > 0$, $W^{-\alpha}\big[e^{-kt}\big](x) = k^{-\alpha}e^{-kx}$.*

Proof. Substitute $u = t - x$ and use $\int_0^\infty u^{\alpha-1}e^{-ku}du = \Gamma(\alpha)k^{-\alpha}$. □

Substituting $t = -x$, Lemma 2.2 is the operator relation used in (1); the limit $x \to 0^-$ there becomes $t \to 0^+$ here.

Theorem 2.3 (the gateway identity)**.** *Let $F_N(t) = \sum_{k=1}^N e^{-kt} = e^{-t}(1-e^{-Nt})/(1-e^{-t})$. For $\mathrm{Re}(\alpha) > 1$,*

$$\lim_{t\to 0^+} \lim_{N\to\infty} (W^{-\alpha}F_N)(t) = \zeta(\alpha),$$

the limits interchangeable, and the identity is equivalent to the classical representation

$$\Gamma(\alpha)\,\zeta(\alpha) = \int_0^\infty \frac{t^{\alpha-1}}{e^t-1}\,dt. \tag{2}$$

Proof. By Lemma 2.2, $(W^{-\alpha}F_N)(t) = \sum_{k\le N} k^{-\alpha}e^{-kt}$. Since $|F_N(t)| \le (e^t-1)^{-1}$ uniformly in N and $(t-x)^{\alpha-1}(e^t-1)^{-1}$ is integrable on (x,∞) for $x>0$, dominated convergence passes $N\to\infty$ through $W^{-\alpha}$; since $\sum_k k^{-\mathrm{Re}\alpha} < \infty$, dominated convergence for series passes $t\to 0^+$ through the sum, giving $\zeta(\alpha)$. Although $B(t) = (e^t-1)^{-1}$ is not locally bounded at 0, for $\mathrm{Re}\alpha > 1$ the integral defining $(W^{-\alpha}B)(0)$ converges absolutely; we use the same notation for this natural extension. Taking $N\to\infty$ first and evaluating at $x=0$ gives $(W^{-\alpha}[(e^t-1)^{-1}])(0) = \Gamma(\alpha)^{-1}\int_0^\infty t^{\alpha-1}(e^t-1)^{-1}dt$, which therefore equals $\zeta(\alpha)$; this is (2), and conversely (2) with Lemma 2.2 recovers the identity. □

Identity (2) is the representation from which Riemann's memoir [21] departs; the fractional construction is thus an independent route to the correct analytic gateway.

Proposition 2.4 (finite base points break the eigenrelation)**.** *Let I_0^α denote the fractional integral of order $\alpha>0$ with base point 0. For $\lambda>0$,*

$$I_0^\alpha\big[e^{\lambda t}\big](x) = \lambda^{-\alpha}e^{\lambda x}\,P(\alpha,\lambda x), \qquad P(\alpha,z) = \frac{\gamma(\alpha,z)}{\Gamma(\alpha)},$$

the regularized lower incomplete gamma function: the eigenrelation of Lemma 2.2 holds only up to the endpoint factor $P(\alpha,\lambda x)$, which tends to 1 as $x\to+\infty$ and to 0 as $x\to 0^+$, and is $\equiv 1$ exactly in the translation-invariant limit (base point $\to-\infty$), i.e. for the Weyl/Liouville operator used in Theorem 2.3.

Proof. $\Gamma(\alpha)^{-1}\int_0^x (x-t)^{\alpha-1}e^{\lambda t}\,dt = e^{\lambda x}\,\Gamma(\alpha)^{-1}\int_0^x u^{\alpha-1}e^{-\lambda u}\,du = \lambda^{-\alpha}e^{\lambda x}\,\gamma(\alpha,\lambda x)/\Gamma(\alpha)$. □

Remark 2.5. The derivative-side statement is the same phenomenon: with base point 0 and non-integer α, $m=\lceil\alpha\rceil$, one has ${}^{RL}D^\alpha[e^{\lambda x}] = x^{-\alpha}E_{1,1-\alpha}(\lambda x)$ and ${}^{C}D^\alpha[e^{\lambda x}] = \lambda^m x^{m-\alpha}E_{1,m-\alpha+1}(\lambda x)$ (Mittag–Leffler functions; see [13]), which diverge, respectively vanish, as $x\to 0^+$, the eigenvalue emerging only asymptotically at infinity. We emphasize the scope of this comparison: it concerns the finite-base-point operators. Other *translation-invariant* fractional operators (the Liouville derivative, the Marchaud derivative) also possess exact exponential eigenrelations under their usual definitions; the dichotomy relevant to the gateway is translation invariance versus endpoint memory, not the Weyl operator versus all others.

3 The strip reformulation, and the limits of symmetry

For $0<\mathrm{Re}(s)<1$ the regularized form of (2) reads $K(s) := \Gamma(s)\zeta(s) = \int_0^\infty x^{s-1}\big((e^x-1)^{-1} - x^{-1}\big)dx$. Substituting $u=\log x$ and $s=\frac12+iz$:

Lemma 3.1. *Let $\Phi(u) = e^{u/2}\big((e^{e^u}-1)^{-1} - e^{-u}\big)$ and $G(z) = \int_{-\infty}^\infty \Phi(u)e^{izu}du$. Then $|\Phi(u)| \ll e^{-|u|/2}$, G is analytic on $|\mathrm{Im}z| < \frac12$ with $G(z) = K(\frac12+iz)$, and, since Γ is zero-free, the non-trivial zeros of ζ correspond bijectively to the zeros of G in the strip under $s=\frac12+iz$.* RH is the statement that all zeros of G in the strip are real.

Proof. The decay is read off the two ends of Φ; analyticity and the identity follow by Fubini–Morera and continuation from the real axis. □

The functional equation and Schwarz reflection make the zero multiset of G invariant under $z \mapsto -z$ and $z \mapsto \bar{z}$. The following elementary observation delimits precisely what those symmetries can ever prove.

Proposition 3.2 (two points, not infinitely many conditions)**.** *A hypothetical off-line zero $z_0 = t + i\delta$ of G ($0 < |\delta| < \frac{1}{2}$) amounts to one complex condition $G(t + i\delta) = 0$ — equivalently two real scalar conditions — the symmetries then forcing the vanishing of G at the remaining members of the quartet $\{\pm t \pm i\delta\}$; and to nothing more. In particular no argument from {functional equation, reflection, kernel decay, analyticity} alone can exclude off-line zeros: the even entire function $q(z) = \big((z-t)^2 + \delta^2\big)\big((z+t)^2 + \delta^2\big)e^{-z^4}$ is real on $\mathbb{R}$, decays rapidly on every fixed horizontal strip, has (for $t \neq 0$) simple zeros at exactly the symmetry quartet $\{\pm t \pm i\delta\}$, and satisfies $q(t) = \delta^2(4t^2 + \delta^2)e^{-t^4} \neq 0$; and, decisively, the Davenport–Heilbronn function [7] satisfies all four hypotheses and possesses infinitely many off-line zeros.*

Principle 3.3 (the litmus test). Any method with the strength to prove RH must distinguish ζ from the Davenport–Heilbronn function, and must therefore consume arithmetic information that Davenport–Heilbronn lacks; the Euler product is the canonical — though not the logically unique — source of such information.

The symmetry and analytic inputs of Section 3 are shared with Davenport–Heilbronn (the kernel of Section 2 is of course specific to ζ). The first statement in this paper that consumes ζ's own arithmetic comes next.

4 The explicit formula and the positivity criteria

Lemma 4.1 (the Euler product's entrance)**.** *For* $\mathrm{Re}(s) > 1$, $-\zeta'/\zeta(s) = \sum_{n\geq 2} \Lambda(n) n^{-s}$, *with* Λ *the von Mangoldt function; the coefficients are* non-negative *and supported on prime powers.*

Proof. Logarithmic differentiation of the absolutely convergent Euler product. □

Pairing test functions against ξ'/ξ by contour integration yields the Riemann–Weil explicit formula (see [26], [14, Ch. 5]): for even h, holomorphic and $O((1+|z|)^{-2-\delta})$ on $|\mathrm{Im} z| \leq \frac{1}{2} + \delta$, with the Fourier convention $g(u) = \frac{1}{2\pi}\int_{-\infty}^{\infty} h(r) e^{-iru}\, dr$,

$$\sum_{z_\rho} h(z_\rho) = h\big(\tfrac{i}{2}\big) + h\big(-\tfrac{i}{2}\big) + \frac{1}{2\pi}\int_{-\infty}^{\infty} h(r)\,\Omega(r)\, dr - 2\sum_{n\geq 2} \frac{\Lambda(n)}{\sqrt{n}}\, g(\log n), \tag{3}$$

where z_ρ runs over the zeros of G in the strip and $\Omega(r) = \mathrm{Re}\,\psi(\frac{1}{4} + \frac{ir}{2}) - \log\pi$. The zero side is spectral data of G; the prime side is a pairing against the non-negative measure of Lemma 4.1. *Weil's criterion* [26] states that RH holds iff the right side of (3) is non-negative for every admissible h that is non-negative on $\mathbb{R}$; the forward direction is immediate from (3) (real zeros, non-negative terms), and the converse — constructing, from an off-line zero, an admissible $h \geq 0$ on $\mathbb{R}$ whose zero-side sum is negative — is where the analytic weight lies: the pair values $h(z_0), h(\bar{z}_0)$ carry no sign constraint, and an ε-shifted Gaussian $h(z) = (z - \bar{z}_0 - i\varepsilon)(z - z_0 + i\varepsilon)e^{-2a(z-\gamma)^2}$ makes them $-\varepsilon(2\mu - \varepsilon)e^{2a\mu^2}$ each (one works with the even symmetrization $h(z) + h(-z)$, which is admissible and non-negative on $\mathbb{R}$; symmetrizing adds the reflected value $h(-z_0)$, which for the

selected quartet — whose real part is nonzero, ζ having no real non-trivial zeros — is exponentially negligible against the main pair values, the domination over all remaining zeros being the content of the classical converse); the residual difficulty is dominating all *other* hypothetical off-line zeros, whose heights may approach their supremum without attaining it. We refer to [26, 2, 14] for the classical resolution and turn instead to the discrete criterion, where the difficulty provably disappears.

4.1 Li's criterion, proved

Definition 4.2. With Z the *multiset* of zeros of G in the strip, counted with multiplicity — as are all zero sums in this paper — put, for $z \in Z$, $m(z) = \dfrac{iz-1/2}{iz+1/2}$ (equal to $1-1/\rho$ under $\rho = \frac{1}{2}+iz$), and $\lambda_n = \sum_{z\in Z}^{*}\big(1-m(z)^n\big)$, $n \geq 1$, the sum in symmetric grouping.

Lemma 4.3. *For $z = x+iy$ in the strip: $|m(z)| = 1 \iff y = 0$; $|m(z)| > 1 \iff y > 0$; the symmetry quadruple $\{z, \bar z, -z, -\bar z\}$ maps to $\{m, \bar m, 1/m, 1/\bar m\}$; and $|m(z)|^2 = 1 + \dfrac{2y}{x^2+(1/2-y)^2}$, so that for every $\varepsilon > 0$ only finitely many $z \in Z$ have $|m(z)| \geq 1+\varepsilon$.*

Proof. $|iz-\frac{1}{2}|^2 = x^2+(y+\frac{1}{2})^2$ and $|iz+\frac{1}{2}|^2 = x^2+(\frac{1}{2}-y)^2$ give the dichotomies and the modulus formula; $m(z)m(-z) = 1$ and $m(-\bar z) = \overline{m(z)}$ are direct; $|m| \geq 1+\varepsilon$ with $y < \frac{1}{2}$ confines z to a bounded region, which contains finitely many zeros. □

Lemma 4.4. *λ_n is well defined and real; the real zeros contribute, per symmetric pair, $2(1-\cos n\theta) \in [0,4]$ with $\theta = \arg m$, and their total is $O(n^2)$, by the exact identity $|m(z)-1| = |iz+\frac{1}{2}|^{-1}$ (from $m-1 = -(iz+\frac{1}{2})^{-1}$), which equals $(z^2+\frac{1}{4})^{-1/2}$ for real z and is $\asymp |z|^{-1}$ on the strip, and the density bound $\sum_z (1+|z|^2)^{-1} < \infty$ from the Hadamard theory [25, Ch. 3]. The grouped sum over off-line quadruples also converges absolutely for each fixed n: with $\theta_j \ll 1/|z_j|$ (from $|m(z)-1| = |iz+\frac{1}{2}|^{-1} \asymp |z_j|^{-1}$) and $c_j = \log|m_j| \ll |z_j|^{-2}$ (from the modulus formula of Lemma 4.3), each quadruple contributes $|Q_j(n)| \leq 4(1-\cos n\theta_j) + 8\sinh^2(nc_j/2) \ll n^2|z_j|^{-2}$ once $|z_j|^2 \geq n$, summable by the density bound, and only finitely many quadruples remain below that threshold. Here and throughout, $\sum^*$ denotes $\lim_{T\to\infty}\sum_{|\mathrm{Re}z|\leq T}$ with the quadruple grouping.*

Theorem 4.5 (Li's criterion [18])**.** *RH $\iff$ $\lambda_n \geq 0$ for all $n \geq 1$.*

Proof. ($\Rightarrow$) Under RH all $z \in Z$ are real and Lemma 4.4 exhibits λ_n as a sum of non-negative pair terms.

($\Leftarrow$) Suppose off-line zeros exist. Group them in quadruples with representative modulus $r_j = |m_j| > 1$ and angle θ_j; each quadruple contributes $Q_j(n) = 4 - 2(r_j^n + r_j^{-n})\cos n\theta_j$. By Lemma 4.3 the maximum $\bar r = \max_j r_j$ exists and is attained by finitely many quadruples $j \leq m$, and every non-maximal modulus is uniformly bounded by some $r' < \bar r$: let $S = \{j : r_j < \bar r,\ r_j > (1+\bar r)/2\}$, a finite set by Lemma 4.3, and put $r' = \max\big((1+\bar r)/2,\ \max_{j\in S} r_j\big)$, omitting the second maximum if S is empty; then $r' < \bar r$ and every non-maximal modulus satisfies $r_j \leq r'$. For the non-extremal quadruples write $c_j = \log r_j$ and split $|Q_j(n)| \leq 4(1-\cos n\theta_j) + 8\sinh^2(nc_j/2)$: for $|z_j| \geq \sqrt{n}$ one has $c_j \ll |z_j|^{-2}$, whence $\sinh^2(nc_j/2) \ll n^2|z_j|^{-4}$, summable to $O(n^2)$; the finitely many with $|z_j| < \sqrt{n}$ number $O(\sqrt{n}\log n)$ and are each $\leq 8r'^n$. By simultaneous Dirichlet approximation there are infinitely many n with $\cos n\theta_j \geq \frac{1}{2}$ for all $j \leq m$ simultaneously; along that subsequence

$$\lambda_n \ \leq\ Cn^2 + Cn^{1/2}(\log n)\, r'^n + m\big(4-\bar r^{\,n}\big) \ \longrightarrow\ -\infty,$$

so some $\lambda_n < 0$. □

Remark 4.6. The proof's decisive step is Lemma 4.3: the Cayley map compresses "real part $\to \infty$" into "modulus $\to 1$", so extremal moduli are attained by finitely many zeros — the non-attainment pathology that burdens Weil-type converses cannot be *expressed* in the disc. The obstacle is dissolved by a change of geometry, not defeated by force. A third, arithmetic resolution of the same obstacle — Landau's non-negativity/singularity mechanism, which never selects an extremal zero at all — underlies the Robin–Lagarias channel of Section 8; in all three resolutions the load-bearing wall is a positivity.

Corollary 4.7. *The proof of the converse used only the quadruple symmetry, the density bound, and Lemma 4.3, all of which hold for the Davenport–Heilbronn* critical-strip *zero multiset; since that function has off-line zeros [7], some of the Li-type coefficients formed from its critical-strip zero multiset are negative, unconditionally. (Davenport–Heilbronn may have zeros outside the strip as well; those play no part here.) The family $\{\lambda_n \geq 0\}$ thus passes Principle 3.3: it provably characterizes RH for ζ and provably fails for the counterexample. (Its arithmetic expression through Λ is due to Bombieri–Lagarias [1].)*

Corollary 4.8. *Extend λ to $\mathbb{Z}$ by $\lambda_0 = 0$, $\lambda_{-n} = \lambda_n$. Then RH holds iff the kernel $(j,k) \mapsto \lambda_{j-k}$ is conditionally negative definite. Indeed, under RH, for $\sum c_j = 0$, using $\cos((j-k)\theta) = \frac{1}{2}\big(e^{i(j-k)\theta} + e^{-i(j-k)\theta}\big)$ and the factorization of each exponential double sum as a modulus square,*

$$\sum_{j,k} c_j\,\bar{c}_k\big(1 - \cos((j-k)\theta)\big) = -\tfrac{1}{2}\Big(\Big|\sum_j c_j e^{ij\theta}\Big|^2 + \Big|\sum_j c_j e^{-ij\theta}\Big|^2\Big) \;\leq\; 0$$

pointwise in θ (for complex coefficients both modulus-square terms occur and are in general distinct); integrating against the Lévy–Khinchine representation furnished by Lemma 4.4 gives conditional negative definiteness. Conversely $c_0 = 1$, $c_n = -1$ forces $\lambda_n \geq 0$ for all n, and Theorem 4.5 applies. (Presumably known in substance; we have not located a reference.)

5 The semigroup is the Mellin translation flow

The one algebraic structure the fractional formulation possesses beyond the classical one is the semigroup law $W^{-\alpha}W^{-\beta} = W^{-(\alpha+\beta)}$. It is natural to ask whether it imposes constraints invisible classically. The computations below identify the structure precisely; a discussion of what they do and do not establish follows them.

Theorem 5.1. *Let $\mathrm{Re}\beta > 0$. Although $B(t) = (e^t - 1)^{-1} \sim 1/t$ at $t = 0$, and therefore lies outside the class of Definition 2.1, for every fixed $x > 0$ the defining Weyl integral converges absolutely; all operator identities in this theorem are asserted for $x > 0$. Applied to B, the semigroup implements exactly the translation flow $s \mapsto s + \beta$ on ζ: $W^{-\beta}B = \mathrm{Li}_\beta(e^{-x})$ for $x > 0$, with Mellin transform $\Gamma(s)\zeta(s+\beta)$ on $\mathrm{Re}\,s > 0$, $\mathrm{Re}(s+\beta) > 1$, and by analytic continuation elsewhere. Consequently, for real $\beta > 0$ (to which every use below pertains; for complex β one must in addition avoid the countable exceptional set where a translated zero meets a pole of Γ, or where β itself is a zero, so that $\sigma_j(\beta)$ below is defined):* (i) *the flow transports the non-trivial zeta-zero multiset rigidly, $\{\rho\} \mapsto \{\rho - \beta\}$;* (ii) *for real $\beta \neq 0$ no function whose non-trivial zero multiset is* exactly *the translated multiset can satisfy a functional equation of Riemann type $s \leftrightarrow 1 - s$ (two distinct axes of symmetry would generate an unbounded horizontal orbit*

of zeros inside a bounded strip); (iii) *the full induced structure on the Li data is the classical power-sum calculus* $\lambda_n = \sum_{j\le n}(-1)^{j+1}\binom{n}{j}\sigma_j$, $\frac{d}{d\beta}\sigma_j = j\sigma_{j+1}$ *(with* $\sigma_j = \sum^* \rho^{-j}$ *and, along the flow,* $\sigma_j(\beta) := \sum^*(\rho-\beta)^{-j}$, $\lambda_n(\beta)$ *formed from the translated multiset).*

Proof. $W^{-\beta}[e^{-kt}] = k^{-\beta}e^{-kt}$ termwise gives the polylogarithm; its Mellin transform is computed termwise. (i) is the definition; (ii) is the two-axes argument stated; (iii) is the binomial expansion (grouped convergence by pairing ρ with $1-\rho$) together with differentiation of $\sigma_j(\beta) = \sum^*(\rho-\beta)^{-j}$ under the sum. □

Remark 5.2 (what these computations do and do not establish). They identify the fractional semigroup with familiar Mellin and polylogarithmic structure: a flow that transports zeros rigidly, leaves the functional-equation class instantly, and reproduces on the Li data exactly the classical power-sum relations. They support — but, absent a formal definition of the class of admissible deductions, do not constitute a theorem asserting — the qualified conclusion that the semigroup offers no evident leverage on zero location: heuristically, a sign constraint obtained from the flow alone would be translation-equivariant and would apply to translated, non-symmetric configurations as well; Corollary 4.7 provides a separate illustration that positivity can fail for zero configurations lacking ζ's arithmetic (the Davenport–Heilbronn multiset is not itself a translated zeta configuration). We state this as an assessment, not a theorem.

The hoped-for pun between $\alpha+\beta$ in the semigroup and $m+n$ in the Li indices does not survive inspection: under the Cayley map the semigroup is a parabolic Möbius flow fixing the boundary point $m=1$, while the Li indexing is the power map. Positivity lives entirely in the second. The gateway of Section 2 is thereby evaluated: correct as a representation; and, so far as these computations reach, offering no lever beyond the classical structure they reproduce.

6 The estimation ladder

The estimation theory enters through the incomplete-gamma structure implicit in Proposition 2.4: a finite base point is a truncation, its exact price is an incomplete gamma factor, and Riemann's theta-smoothed representation

$$\pi^{-s/2}\Gamma\big(\tfrac{s}{2}\big)\zeta(s) = \frac{1}{s(s-1)} + \int_1^\infty \big(x^{s/2-1} + x^{(1-s)/2-1}\big)\omega(x)\,dx, \qquad \omega(x) = \sum_{n\ge1} e^{-\pi n^2 x},$$

is the split for which the remainder is symmetric. From it descend, by classical arguments we only cite here: Hardy's theorem (infinitely many zeros on the line) [9, 25]; the balanced approximate functional equation of Hardy–Littlewood with lengths $\sqrt{t/2\pi}$ [11, 25]; the convexity bound $\zeta(\frac12+it) \ll t^{1/4+\varepsilon}$ (current record $t^{13/84+\varepsilon}$, Bourgain [4]); the second moment $\int_0^T |\zeta(\frac12+it)|^2 dt = T\log\frac{T}{2\pi} + (2\gamma-1)T + E(T)$ [10, 12]; and Ingham's fourth moment $\sim \frac{1}{2\pi^2}T\log^4 T$ [12], whose off-diagonal is governed by the shifted divisor sums — the exact point past which (at the sixth moment, with d_3-correlations) the ladder becomes conjectural [19, 15]. One remark connects the two halves of this paper: the phase that governs every cross term in the mean-value theory has derivative $\vartheta'(t) = \frac12\Omega(t)$ — *the archimedean weight of the explicit formula* (3) *meters the moments and, through* $N(T) = \vartheta(T)/\pi + O(\log T)$, *counts the zeros.* Reformulation and estimation run on one engine.

We give one diagnostic and prove one theorem: an observation locating the necessity behind the architecture of the density and proportion results, and a zero count on the line.

Remark 6.1 (why naïve winding cannot count critical-line zeros). The functional equation forces every natural candidate for a "winding function" on the critical line to be *real-valued* there: $\xi(\frac{1}{2}+it) = \Xi(t)$ and Hardy's $Z(t) = e^{i\vartheta(t)}\zeta(\frac{1}{2}+it)$ are both real. A real-valued trace carries sign-change information, not winding information: any attempt to credit the zero count with the full argument growth $\vartheta(T) \sim \pi N(T)$ of the archimedean factor founders on the fact that, along the line, this growth is offset inside the degenerate argument of the real trace, the offset being supplied precisely by the zeros — including any hypothetical off-line ones, which such an argument cannot see separately. We therefore make no quantitative claim here; the qualitative conclusion is this: the reality of Ξ or Z alone cannot convert the full argument growth into a count of critical-line zeros. A *winding-based comparison with* $N(T)$ therefore needs information separating on-line from off-line zeros — the rôle of Littlewood's lemma applied to mollified mean values on a line $\sigma_0 = \frac{1}{2} - R/\log T$, and — since $|\zeta|$ is inflated by e^R-factors there — of Levinson's derivative combination, engineered to cancel the inflation; this is the architecture of [17, 6, 5], read as necessity. Direct sign-change methods, such as Hardy's method and Theorem 6.2 below, provide a different route, requiring no off-line control at all.

A different, vertical-line route — classical unmollified mean values together with the Littlewood–Jensen mechanism — yields the Bohr–Landau theorem (for every $\varepsilon > 0$, all but $o(N(T))$ zeros satisfy $|\beta - \frac{1}{2}| < \varepsilon$) [3]; mollified mean values and derivative combinations, by contrast, are the engine of the Levinson–Conrey proportion method; on the zero-density side, the recent large-values breakthrough of Guth–Maynard [8] gives $N(\sigma, T) \le T^{30(1-\sigma)/13+o(1)}$, the first improvement near $\sigma = \frac{3}{4}$ on Ingham's exponent in decades; and Levinson's method yields the proportion ladder $\kappa \ge 0.3474$ [17], $\kappa \ge 2/5$ [5] (mollifiers past length $T^{1/2}$ via Kloosterman sums), and currently $\kappa > 5/12$ [20]. Below that industrial altitude, the following is what the approximate functional equation and the two moment theorems yield by themselves.

Theorem 6.2. *Let $N_0(T)$ count the zeros of ζ on the critical line with $0 < \gamma \le T$. Then $N_0(T) \gg T^{3/4}/\log^5 T$.*

Proof. Let $Z(t) = e^{i\vartheta(t)}\zeta(\frac{1}{2}+it)$, real-valued with $|Z| = |\zeta(\frac{1}{2}+it)|$, and (from the balanced AFE and $\chi = e^{-2i\vartheta}$) $Z(t) = 2\sum_{n\le\sqrt{t/2\pi}} n^{-1/2}\cos(\vartheta(t) - t\log n) + O(t^{-1/4})$.

Signed area is small. For $T \le t \le 2T$ and $0 < H \le T^{1/4}\log^3 T$, freeze the cutoff at $M = \lfloor\sqrt{t/2\pi}\rfloor$ (a symbol chosen to avoid collision with the zero-counting function N_0); the AFE error $O(u^{-1/4})$ integrates to $O(HT^{-1/4}) = O(\log^3 T)$, and the $O(1 + HT^{-1/2})$ terms entering or leaving the sum as the cutoff drifts contribute $O\big(H(1+HT^{-1/2})T^{-1/4}\big) = O(\log^3 T)$ as well; both are subordinate to the bound claimed below. The phase $p_n(u) = \vartheta(u) - u\log n$ has $p_n'(u) = \log(\sqrt{u/2\pi}/n) + O(u^{-2})$ and $p_n'' \asymp 1/u > 0$. The first derivative test gives $O(1)$ per term for $n \le M/2$, and with the weights $n^{-1/2}$ the total is $\ll \sum_{n\le M} n^{-1/2} \ll T^{1/4}$; in the near range it gives $O(M/(M-n))$ per term, total $\ll M^{-1/2}\sum_{1\le r\le M/2} M/r \ll T^{1/4}\log T$; the $O(1)$ remaining terms in the stationary band $p_n' < T^{-1/2}$ (forcing $M - n = O(1)$) fall to the second derivative test with $p'' \asymp 1/T$, contributing $\ll M^{-1/2}\sqrt{T} \ll T^{1/4}$ in total. Hence $\big|\int_t^{t+H} Z\big| \ll T^{1/4}\log T$, uniformly in such H.

Unsigned mass is large. By Hölder, $\int_T^{2T}|Z| \ge \big(\int Z^2\big)^{3/2}\big(\int Z^4\big)^{-1/2} \gg (T\log T)^{3/2}(T\log^4 T)^{-1/2} = T\log^{-1/2} T$, by the second and fourth moments [10, 12].

Count. Partition $[T, 2T]$ into half-open intervals $[t, t+H)$ of length $H = T^{1/4}\log^3 T$, with the initial offset chosen so that no partition endpoint is a zero of Z (possible, the zeros being discrete); then no point is counted twice and every sign change lies inside some interval. The

shifted grid leaves at most two boundary fragments of total length $\le 2H$; by Cauchy–Schwarz their combined unsigned mass is $\ll \big(H\int_T^{2T} Z^2\big)^{1/2} \ll T^{5/8}\log^2 T = o(T\log^{-1/2} T)$, so they may be discarded. Intervals without a sign change of Z have $\int_I |Z| = |\int_I Z| \ll T^{1/4}\log T$ and contribute in total $\ll T/\log^2 T$, so the sign-change intervals carry almost all the mass; if s is their number, Cauchy–Schwarz gives $T\log^{-1/2} T \ll s^{1/2}(H\int_T^{2T} Z^2)^{1/2} \ll s^{1/2}T^{5/8}\log^2 T$, whence $s \gg T^{3/4}/\log^5 T$. Each sign change is an on-line zero; sum dyadically. □

Hardy–Littlewood [11] reached $\gg T$ and Selberg [24] the positive proportion by refining exactly the two generous steps above (phase-adapted spacing; mollified short-interval moments). The gap between the exponent 3/4 and the truth is priced, like every other gap on this ladder, in logarithms and mollifiers.

7 Structural remarks

Six patterns recur across the material often enough to read as facts about the problem rather than about any method. *Rigidity:* exact equivalences — analytic (Lemma 3.1), spectral (Theorem 4.5), arithmetic (Section 8) — transport the difficulty intact; there is no softer face of this summit. *Zero margin:* Gronwall's exact e^γ for Robin's bound, and the de Bruijn–Newman constant, for which Rodgers–Tao proved $\Lambda_{\mathrm{dBN}} \ge 0$ [23] while RH would imply $\Lambda_{\mathrm{dBN}} \le 0$ (de Bruijn), hence equality under RH, show RH holding — if it holds — with no room to spare, which is the structural reason symmetry-only arguments (Proposition 3.2) cannot succeed: nothing generic pushes zeros onto a line. *Positivity as the primes' doorway:* the Euler product enters analysis as $\Lambda \ge 0$ (Lemma 4.1), and every working converse mechanism — disc compactification, Landau's singularity argument, the classical Weil apparatus — rests on a positivity; the same lever at partial strength proves the Prime Number Theorem. *One archimedean engine:* Ω weights (3), counts zeros, and meters the moments. *The litmus:* the distance between rhetoric and mathematics on this problem is the distance between claims that cannot distinguish ζ from Davenport–Heilbronn and criteria (Corollary 4.7) that provably do. *The frontier's location:* the technical walls met above are not artifacts of exposition; they are where the subject's open frontier runs — the limits of currently controllable long mollifiers (Conrey's Kloosterman-sum technology having pushed past length $T^{1/2}$) and the unresolved higher divisor correlations.

8 Coda

Robin [22] proved that RH is equivalent to $\sigma(n) < e^\gamma n\log\log n$ for $n \ge 5041$, through the extremal theory of colossally abundant numbers — arithmetic test functions, constructed from primes up to a favorable bound, that convert prime-count fluctuations into divisor-sum excesses; Lagarias [16] removed the constant and the threshold. The Hypothesis whose analytic anatomy has occupied this paper is, exactly, the statement that for every $n \ge 1$,

$$\sum_{d\mid n} d \;\le\; H_n + e^{H_n}\log H_n, \qquad H_n = \sum_{j=1}^{n}\frac{1}{j},$$

with equality only for $n = 1$ — checkable by anyone for any given n, and open. The fractional gateway of (1) does not prove it; no honest gateway yet has. What the gateway does, pursued rigorously, is traverse the subject entire and arrive at the same frontier where the classical authors stand, with the map improved.

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